MILITARY SPECIFICATION

ACETYLENE BLACK
(For Use in Ammunition)

This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging for the manufacture of Acetylene Black (100% compressed) for use in Thionyl Chloride ($SOCl_2$) type battery (see 6.3 and 6.4).

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-1218 - ACS Chemicals

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E300 Sampling Industrial Chemicals
ASTM-C571 Chemical Analysis of Carbon and Carbon Ceramic Refractories
ASTM-D2905 Statements of Number of Specimens Required to Determine the Average of a Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The acetylene black (Shawinigan), when examined by an electronmicroscope as specified in 4.6.1, shall show spherical particles (approximately 100 to 1000 Angstroms) arranged in a chain like fashion.

3.2 Properties. The acetylene black shall conform with the requirements listed in Table 1 when tested in accordance with the applicable paragraphs:

TABLE 1
REQUIREMENT FOR ACETYLENE BLACK

Properties	Requirements	Test Paragraph
Bulk density pounds per cubic foot, lb/cu. ft.	11.0 - 14.0	4.6.2
Moisture, percent, Max.	0.20	4.6.3
Ash %, Max.	0.10	4.6.5
Cyclohexane extract, %, Max.	0.10	4.6.7

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Grit, % Max.	0.20	4.6.6
Carbon, % Minimum (Min)	99.5	4.6.8
Absorption, Stiffness (A.S.) value, Min., milliliters per 5 gram (ml/gm)	17	4.6.4

3.3 First Article Inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 454 g of acetylene black obtained by sampling as described in 4.5.1. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production (see 6.3).

4.3.2 Inspections to be Performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality Conformance Inspection.

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4.4.1 Lot formation for continuous process. A lot shall consist of acetylene black produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each inspection lot shall consist of that quantity of acetylene black that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (See MIL-A-48078). The product shall be submitted for inspection in accordance with MIL-STD-105.

4.5 Testing (see 6.2).

PRECAUTION: This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Two, (2) independent and representative (approximately 454 g) samples shall be withdrawn from each lot of Acetylene Black using the procedures as specified in ASTM E300, section 25. The samples shall be tested independently in accordance with the applicable methods given in 4.6. If either sample fails to meet any of the test requirements, the lot shall be rejected. The samples shall be subjected to the tests listed below.

TABLE II

Inspection/Requirement	Defect Classification
Material (see 3.1)	Major
Properties (see 3.2)	Major

4.5.2 Inspection Equipment. The contractor shall use commercial inspection equipment for all tests and examinations specified in para 4.6. He is also charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use.

4.6 Test Methods and Procedures. (See 6.2) The tests to determine compliance with the requirements given in Table I shall be performed on the Acetylene Black samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be reagent grade or ACS Grade in accordance with MIL-STD-1218.

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4.6.1 Material. The acetylene black shall be examined under a electron microscope to determine compliance with 3.1.

4.6.2 Determination of Bulk Density:

Remove any large lumps in the sample being used with a spatula. Add approximately 200 ml of the sample to a tared 250 ml, clean and dry, graduate cylinder which is fitted with a rubber stopper. Tap the bottom of the cylinder on a large rubber stopper 50 times to settle the material. Read the volume (V) and reweigh the cylinder to 01. g (B). Calculate the bulk density as follows:

4.6.2.1 Calculation

$$\text{Bulk density } (g/cm^3) = \frac{(B - A)}{V}$$

$$\text{Bulk density } (lb/cu.ft.) = \text{B.D. } (g/cm^3) \times 62.4$$

Where: B = Wt. of cylinder + stopper + sample, g
A = Wt. of cylinder + stopper
V = Volume of sample after tapping, cm^3

4.6.3 Determination of Moisture.

4.6.3.1 Apparatus:

weighing bottle with ground-in cover
drying oven
desiccator

4.6.3.2 Procedure:

Quantitatively transfer an accurately weighed portion approximately 5 g) of the sample to a tared weighing bottle fitted with a cover. Place the weighing bottle containing the sample, into an oven and heat at $105^{o} \pm 5^{o}C$ until constant weight is obtained. Remove bottle, cover, cool in a dessicator and reweigh.

4.6.3.3 Calculation. Calculate the moisture as follows:

$$\% \text{ moisture} = \frac{(B - C)}{(B - A)} \times 100$$

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A = Weight of bottle + cover, g.
B = Weight of bottle + cover + sample before heating, g.
C = Weight after heating, g.

4.6.4 Determination of Abosrption - Stiffness (A.S.) Value

4.6.4.1 Apparatus:

500 cm^3 Erlenmeyer flask
500 cm^3 burette

4.6.4.2 Reagents:

10% acetone in water solution (See 4.6)

4.6.4.3 Procedure:

Transfer a 5.0 g portion of the sample into a 500 ml conical flask. The acetone solution is added in the following manner. Initially add 10 ml and shake for 20 seconds, then add 2 ml and shake for 10 seconds. 2 ml additions and 10 seconds shaking are continued until small pellets having a diameter of 1/8" (0.3 cm) to 1/4" (0.6 cm) are formed. At this point the addition of liquid is cut to 1 ml. and continued until the pellets grow to 1/2" (1.3 cm) diameter. The additions are then again cut down to 1 ml. until finally over 2 or 3 spheres are present. From here 1/2 ml additions are used until one sphere is formed. This sphere, however, at first is fragile and will break on shaking. The solution is now added, at 2 drops at a time until the sphere stands fairly vigorous shaking without breaking up. This is the end point. Read the burette. It will be found that a few drops beyond this point, smearing of black on the walls of the flask appears. Report the number of ml. required to form the ball for a 5.0 g sample but before the walls of the flask are smeared.

4.6.5 Determination of Ash

4.6.5.1 Apparatus:

procelain crucible
muffle furnace
desiccator

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4.6.5.2 Procedure:

Weigh accurately a 10 ± 0.1g portion of the sample into a tared porcelain crucible. Place crucible with contents in a muffle furnace maintained at 950° ± 50°C until the sample is completely ignited as evidenced by constant weight. Remove crucible from muffle furnace, cool in a dessicator and weigh.

4.6.5.3 Calculation:

$$\% \text{ ASH} = \frac{\text{g of residue x 100}}{\text{g of sample of black}}$$

4.6.6 Determination of Grit

4.6.6.1 Apparatus:

See Figure 1
drying oven

4.6.6.2 Reagents (See 4.6):

acetone

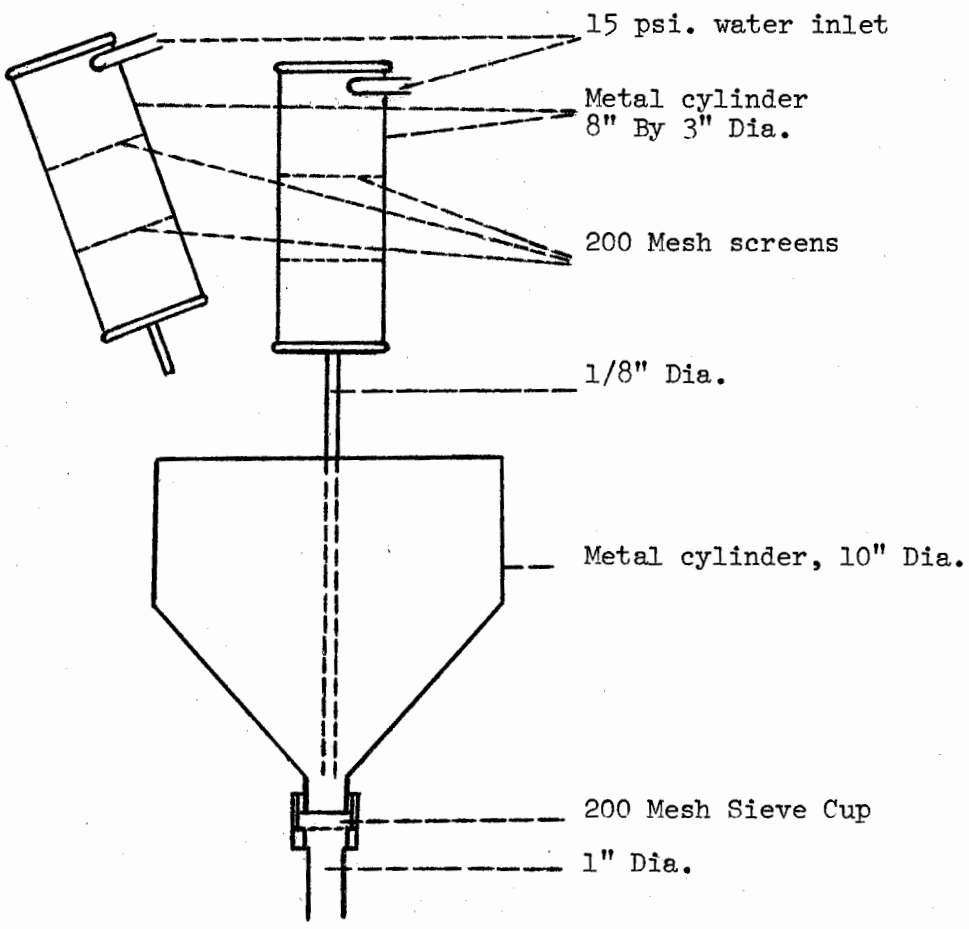
4.6.6.3 Procedure:

Accurately weigh the 200 mash sieve cup (see Figure 1) and install in the apparatus. Quantitatively transfer an accurately weighed portion of the sample, approximately 50 g. to the funnel (see Figure 1). Wet the sample, contained in the funnel, with 100 ml of acetone and allow to filter thru. Then wash the sample with filtered water until the tiltrate is clear. Finally, wash the sample with a 100 ml portion of the acetone. Remove the sieve containing the washed sample, dry at 105° ± 5°C until constant weight is obtained, cool in a dessicator and reweigh.

4.6.6.4 Calculation:

$$\% \text{ GRIT} = \frac{\text{g of residue x 100}}{\text{g of sample}}$$

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APPARATUS FOR
GRIT

FIGURE I

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4.6.7 Determination of Cyclohexane Extract

4.6.7.1 Apparatus:

Soxhlet apparatus
Platinum dish

4.6.7.2 Reagents:

reagent grade cyclohexane

4.6.7.3 Procedure:

Quantitatively transfer an accurately weighed portion of the sample, approximately 10g, into a 43X1233 mm extraction thimble and pack the thimble above the sample with glass wool. Place the thimble containing the sample in a Soxh let apparatus containing cyclohexane and extract for a minimum of 8 hours with a 10 minute cycle. Then, remove thimble, remove glass wool and transfer the extracted sample to a tared platinum dish. Dry the dish and contents in an oven, 60°C maximum, until constant weight is obtained. Remove dish and contents from the oven, cool in a dessicator, and reweigh.

4.6.7.4 Calculation:

$$\% \text{ CYCLOHEXANE EXTRACTABLE} = \frac{\text{g of residue} \times 100}{\text{g of sample}}$$

4.6.8 Carbon. The carbon shall be determined in accordance with ASTM C571 Section 30 to 32.

5. PACKAGING

5.1 Packing. Unless otherwise specified, the Acetylene black shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediate use.

5.2 Marking. Unless otherwise specified, the shipping container shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

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6.1 Ordering data. See MIL-A-48078

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Acetylene black which has been found satisfactory for use in lithium batteries is manufactured by Shawinigan Product Corporation, 111 Charlotte Place, Englewood Cliffs, N.J. Acetylene Black from other sources should be evaluated by ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 before being approved as an equivalent.

6.4 The Acetylene black material is made by thermal decomposition of Acetylene at temperatures above 800°C. It has a more nearly crystalline structure than most Acetylene blacks, hence its unique use as a conductor in batteries.

6.5 The acetylene black is stable indefinitely. Since it is very light and tends to dust easily, personnel should as a precaution use appropriate dust masks when handling.

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